

2025



Case #2

Risk Management

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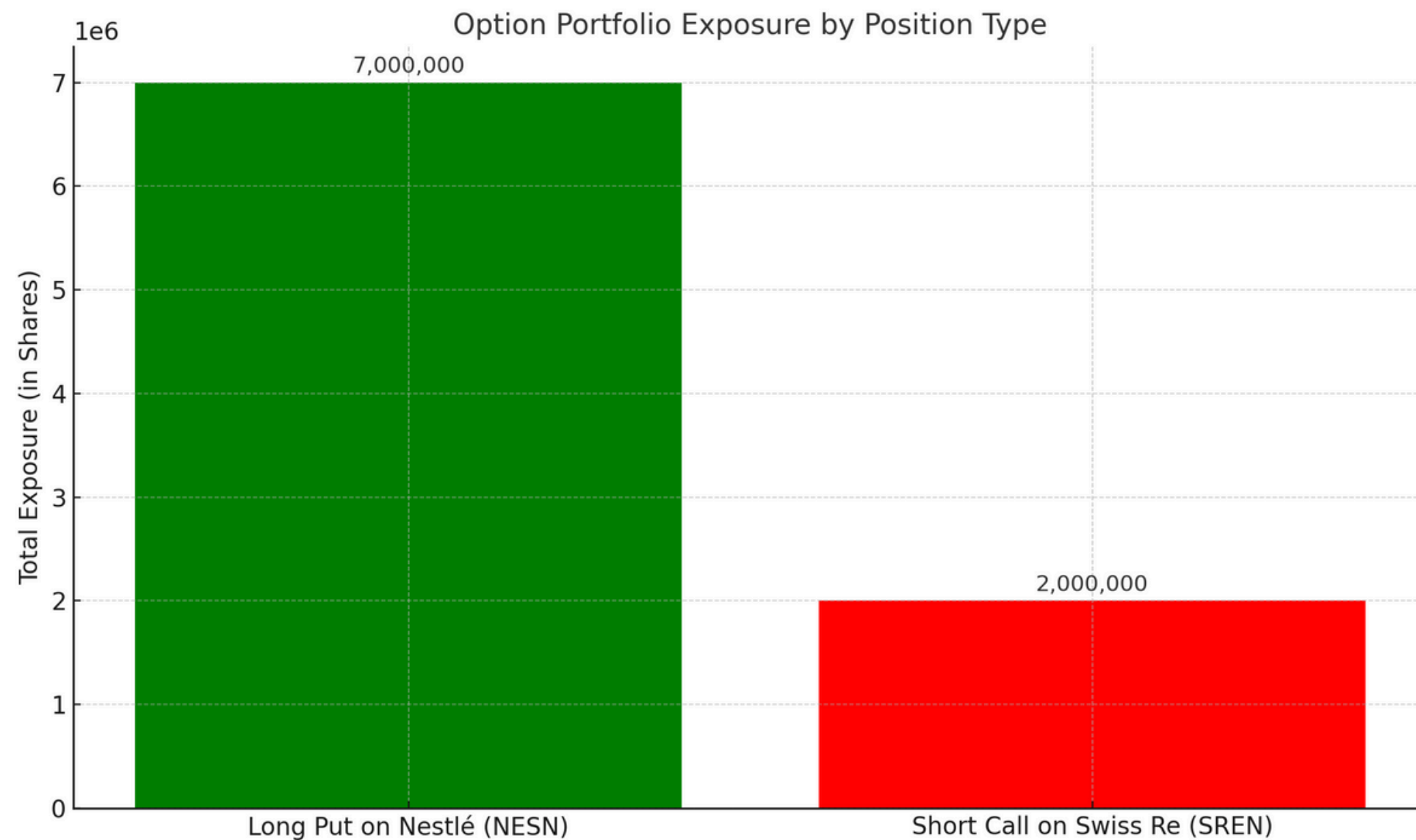
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Part I

Portfolio Description

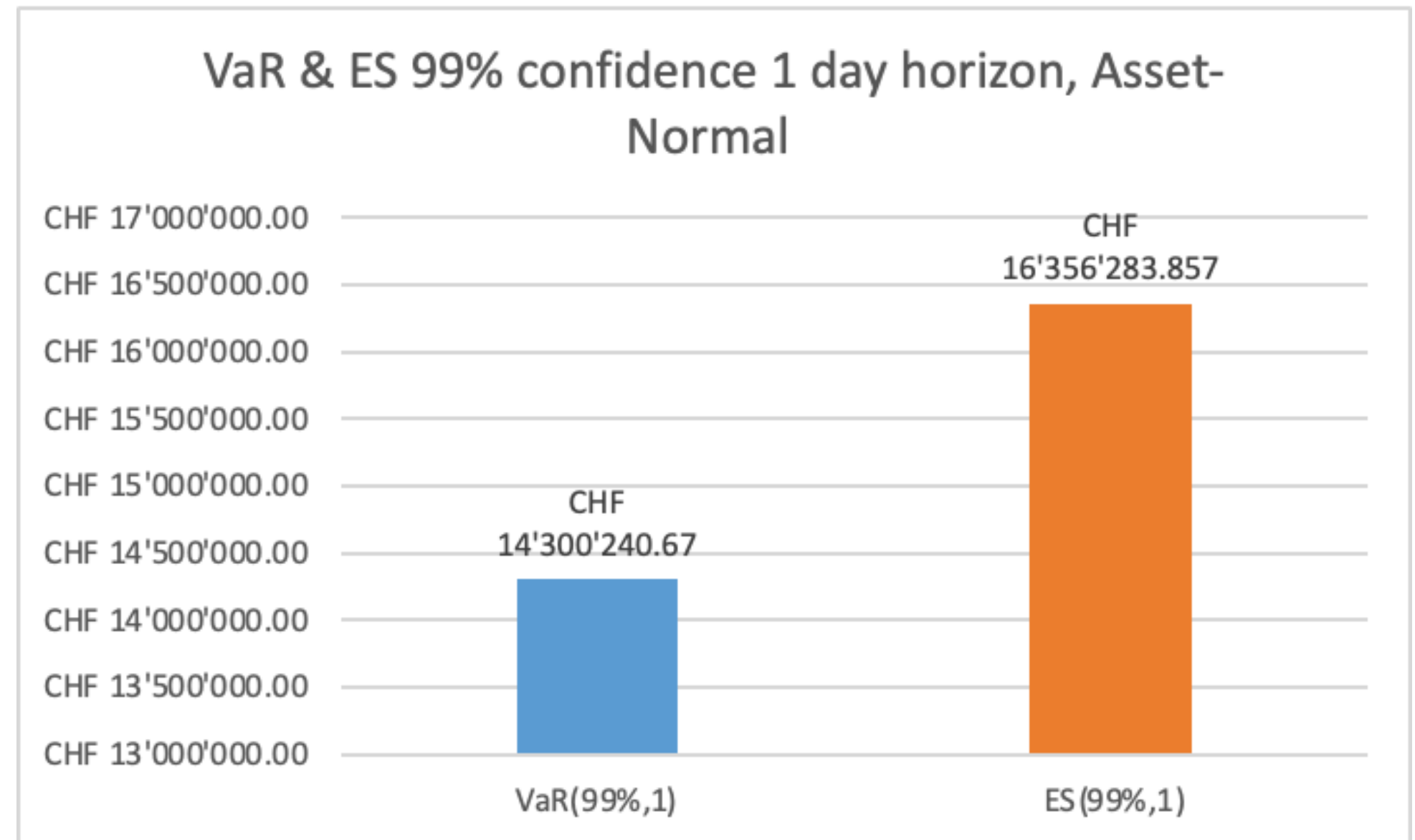


as of 25th of April 2025:

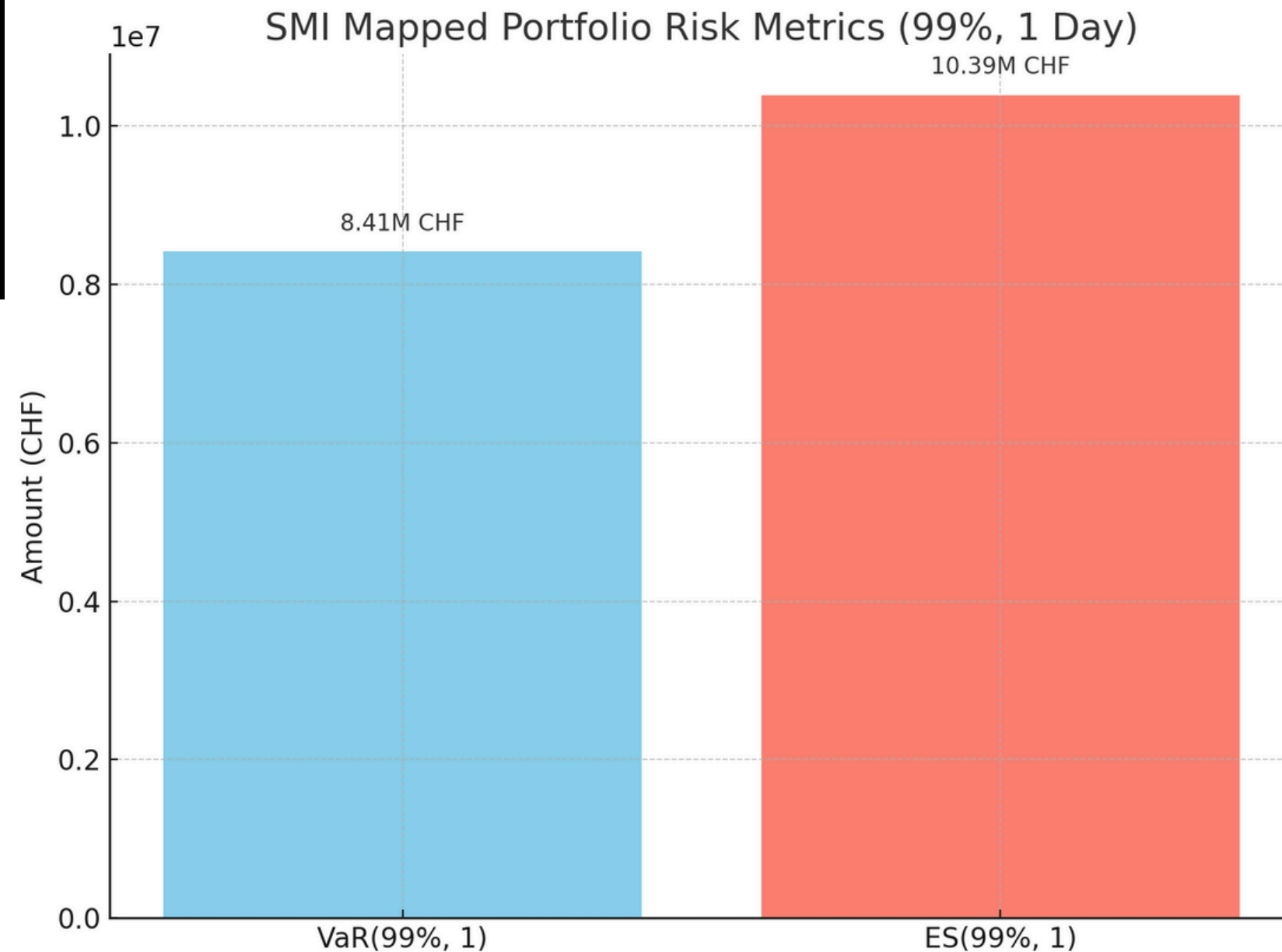
-  Long Position
 - 7,000 Put Options on Nestlé (NESN)
-  Short Position
 - 2,000 Call Options on Swiss Re (SREN)
- Each contract gives exposure to 1,000 shares

Q1: Asset-Normal VaR and ES for Option Portfolio Risk

- The portfolio benefits from diversification, significantly reducing total risk
- Despite high individual exposures, total portfolio risk is lower than the sum of parts.
 - Sum individual VaR: CHF 17'698'763.08
 - Sum individual ES: CHF 47'167'203.62
- Daily portfolio volatility: CHF 6.14M

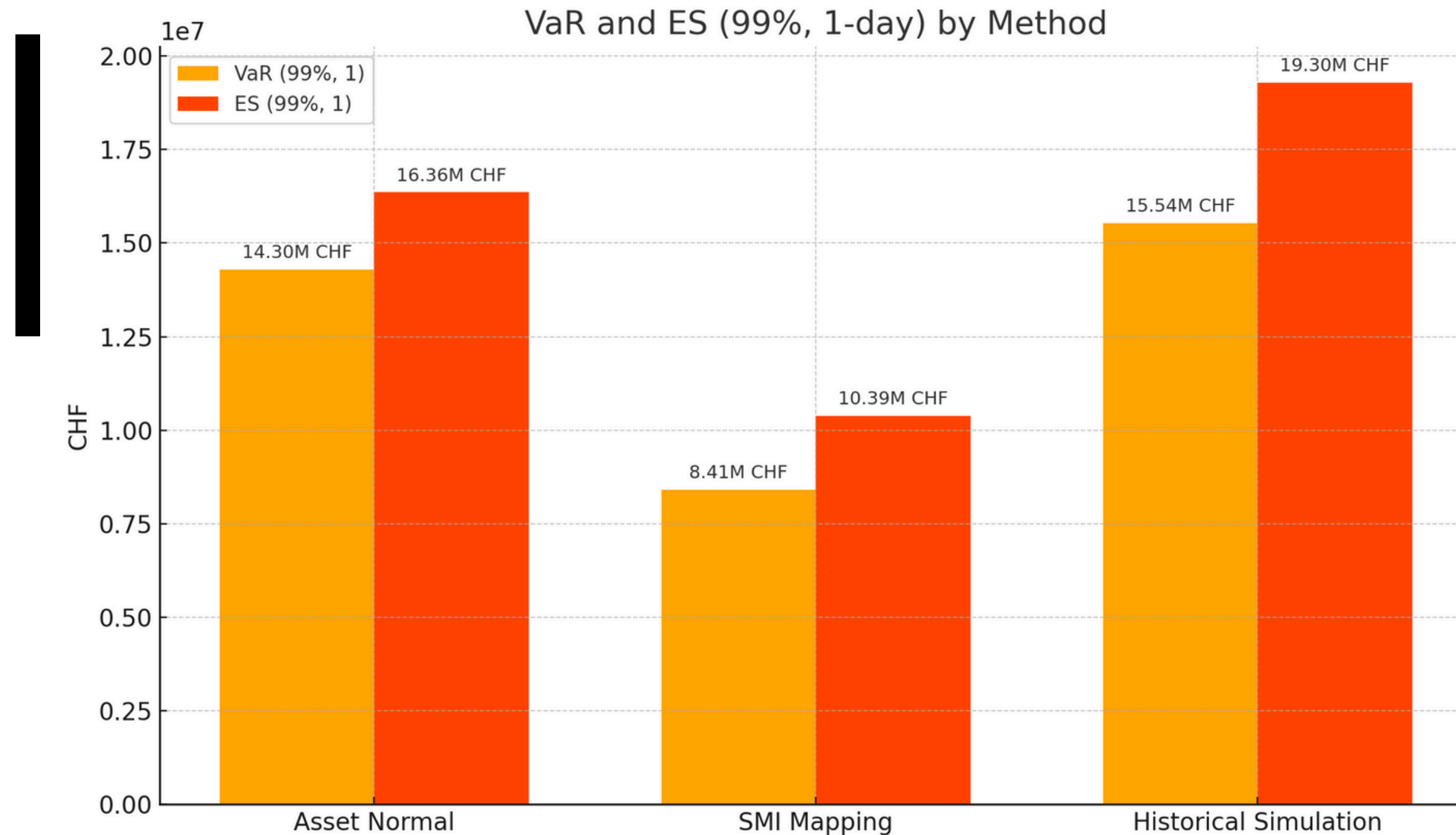


Q2: SMI Beta Mapping Risk Aggregation



- The SMI-Mapped method assumes that all systematic risk is captured via the SMI index, resulting in lower but possibly underestimated risk figures (trade-off between simplicity and completeness).
- Asset-Normal uses actual variance-covariance structure, recognizing that NESN and SREN are only partially correlated ($\rho \approx 0.28$).

Q3: Historical Simulation of Portfolio Risk



- Historical Simulation is most conservative and comprehensive
- Asset-Normal offers a middle-ground between simplicity and accuracy.
- Quick to compute, but with a risk of missing important information in the portfolio.

Part II

Q1: Bond Valuation via Present Discounting

Bond	Quantity	Maturity	Coupon	Price (USD)	Position Value (USD)
1	2,000,000	1 year	0%	96.20	192,400,000
2	-1,500,000	3 years	5%	103.45	-155,175,000
3	4,000,000	5 years	2%	91.62	366,480,000

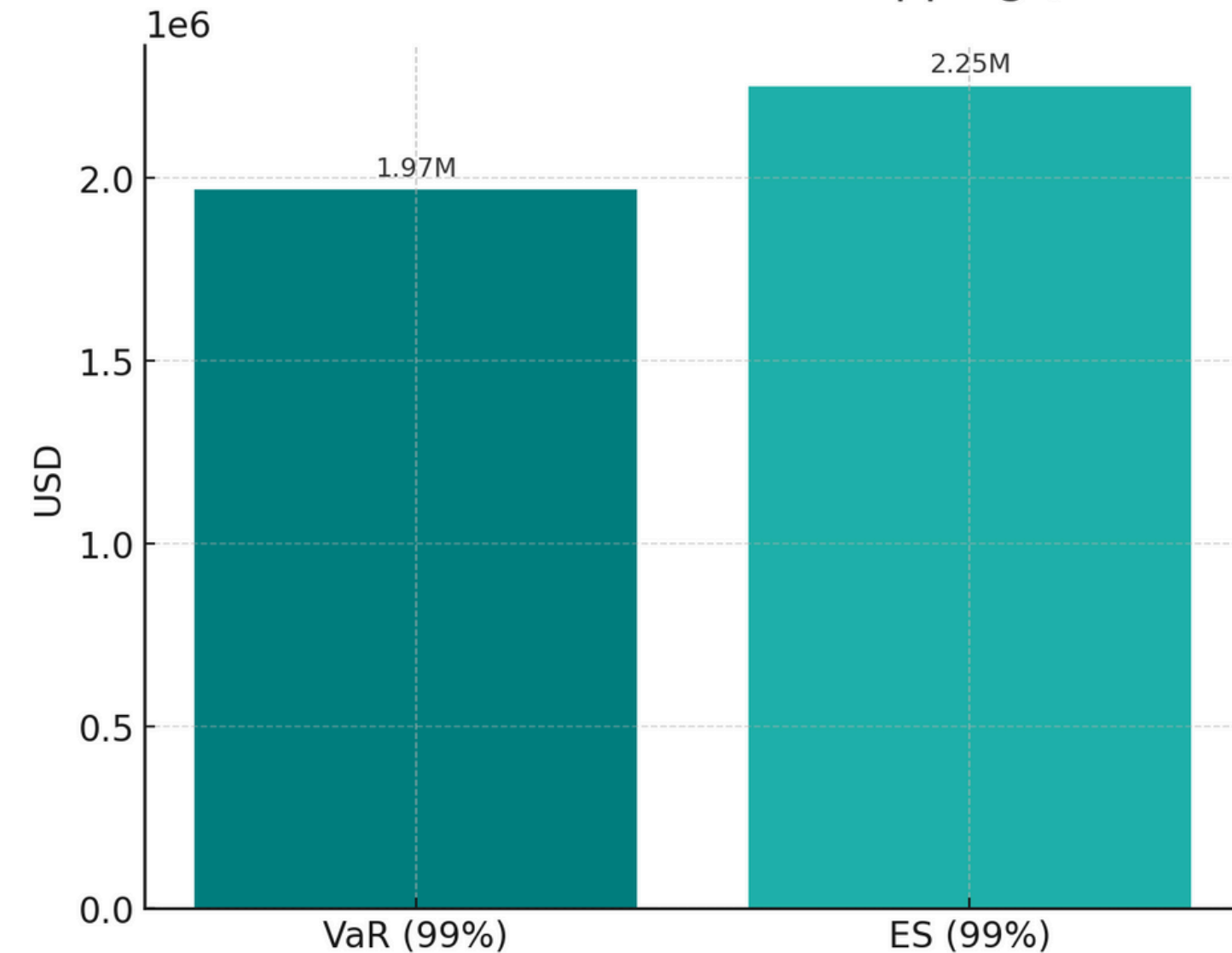
- The short position in Bond 2 (3Y maturity) offsets some of the long exposures
- Portfolio includes a mix of zero-coupon and fixed-coupon instruments, adding duration and interest rate risk variation
- Total Portfolio Value: USD 403,699,002.73
- Pricing Method: Present Value of cash flows discounted using ZCB spot rates (discrete compounding)

Q2: Asset-Normal VaR/ES via Cash-Flow Mapping

Diag V				
0.0004305	0	0	0	0
0	0.0011409	0	0	0
0	0	0.0017356	0	0
0	0	0	0.0018732	0
0	0	0	0	0.0029182

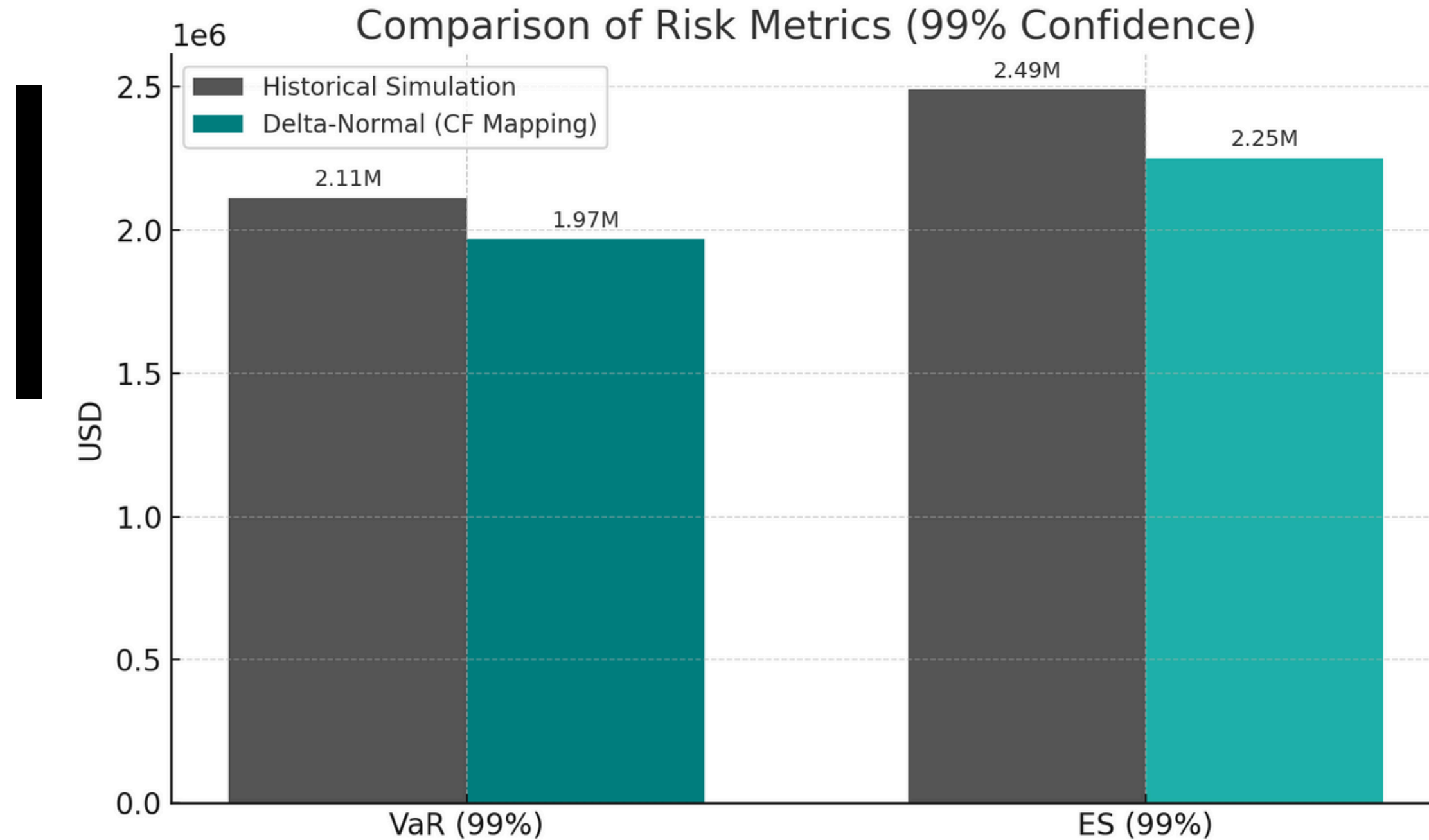
$\Sigma = \text{diag}(V) \times R \times \text{diag}(V)$				
1.853E-07	4.459E-07	6.477E-07	6.592E-07	1.006E-06
4.459E-07	1.302E-06	1.876E-06	1.905E-06	3.028E-06
6.477E-07	1.876E-06	3.012E-06	3.031E-06	4.814E-06
6.592E-07	1.905E-06	3.031E-06	3.509E-06	5.263E-06
1.006E-06	3.028E-06	4.814E-06	5.263E-06	8.516E-06

Asset Normal VaR & ES via Cash-Flow Mapping (99% Confidence)



- The results show that under a 99% confidence level:
- The maximum expected 1-day loss is USD 1.97 million.
- In the worst 1% of cases, the average loss may go up to USD 2.25 million.

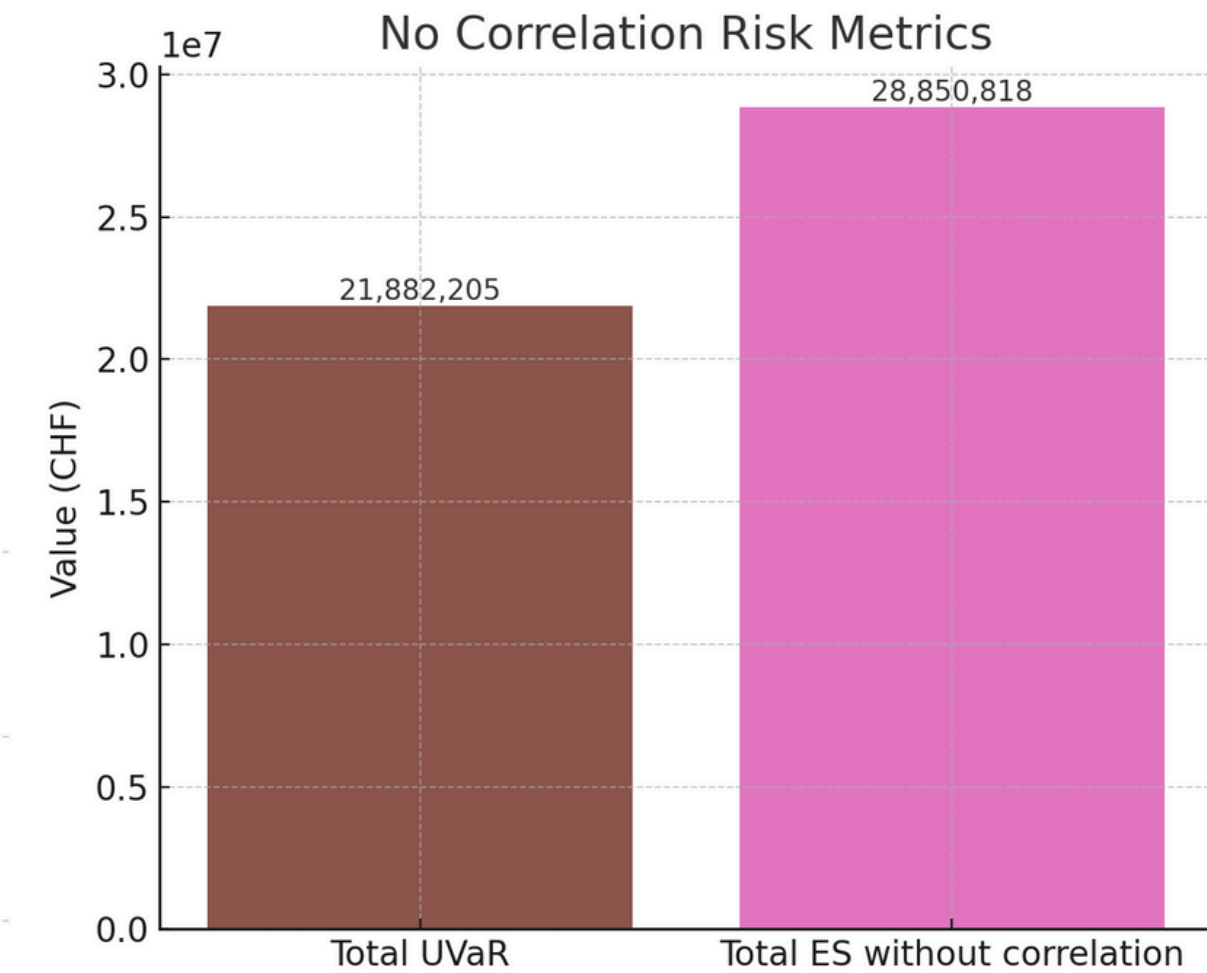
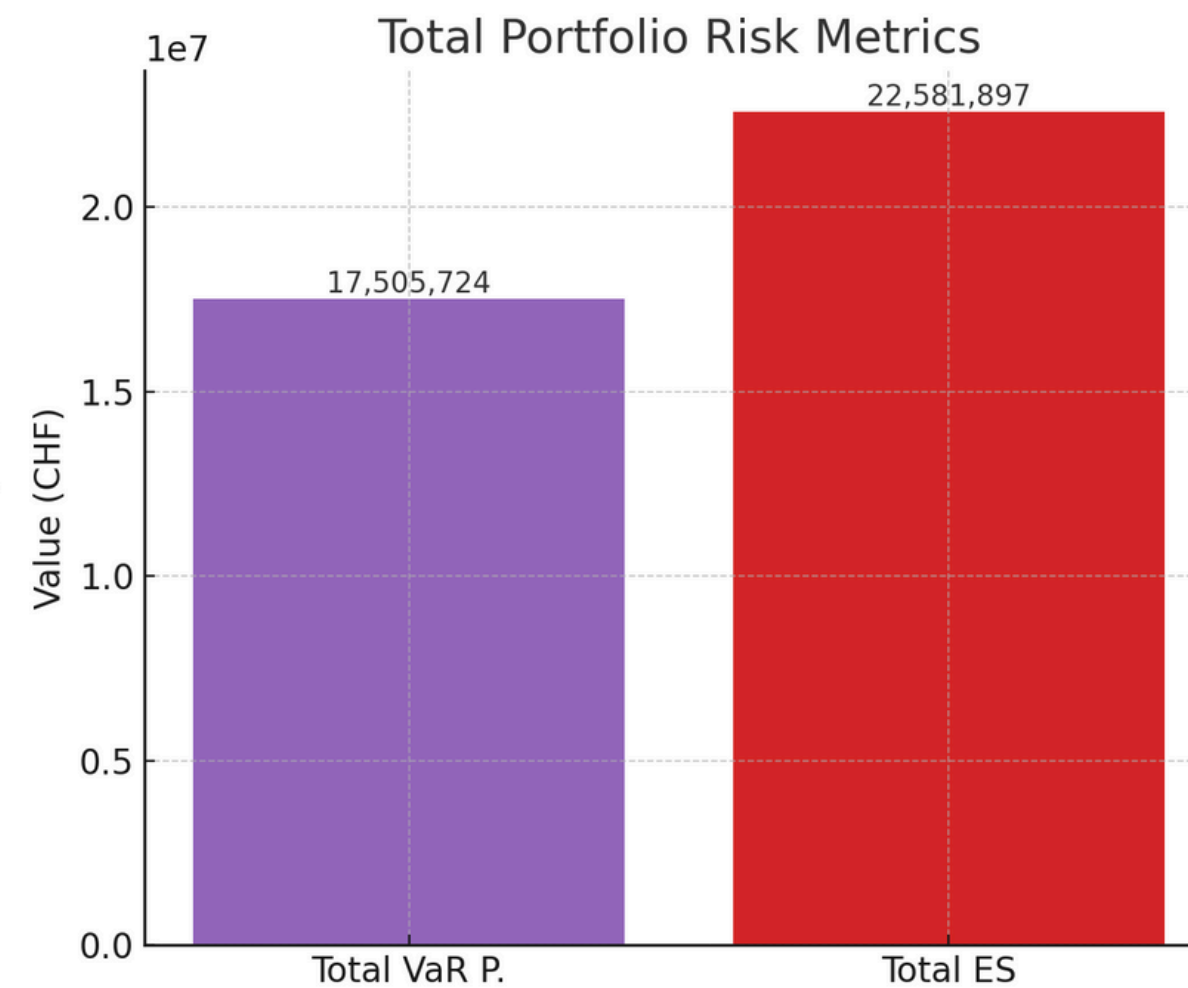
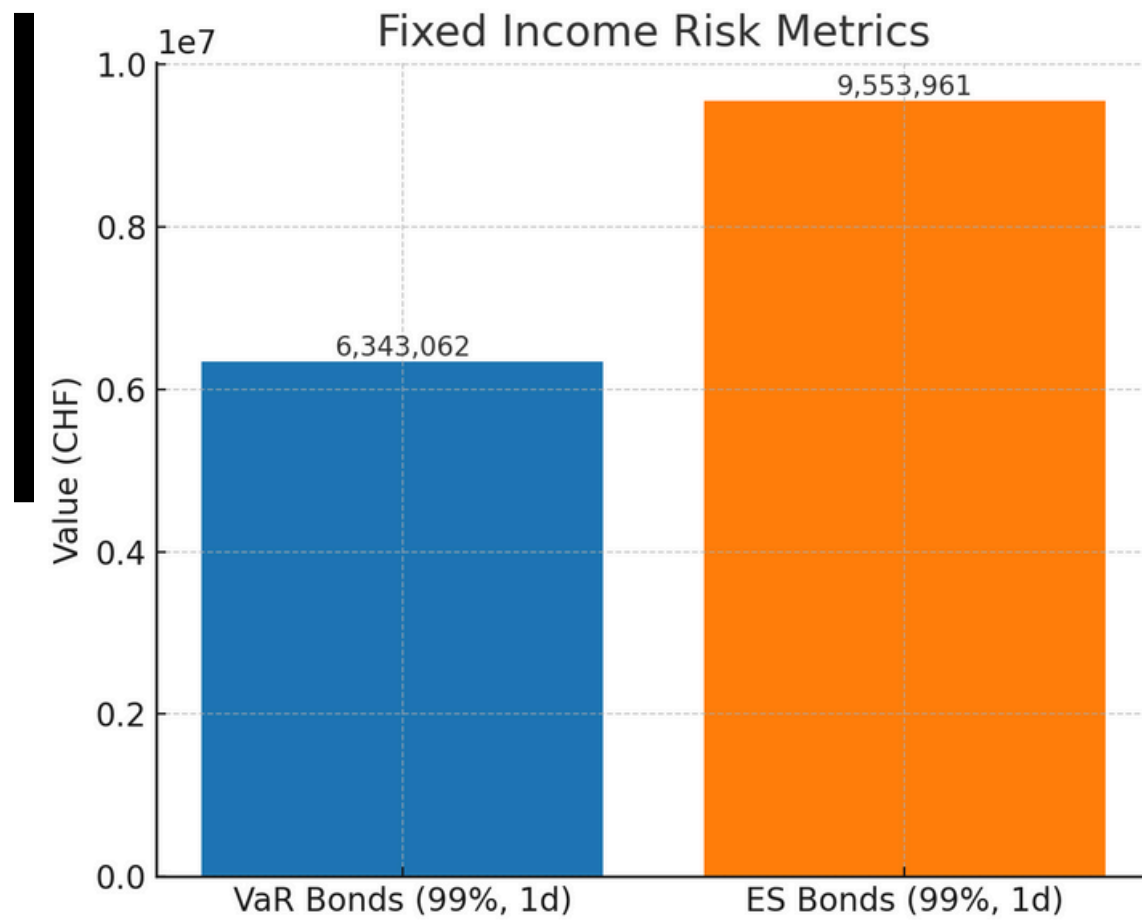
Q3: Historical Simulation for Fixed Income Risk



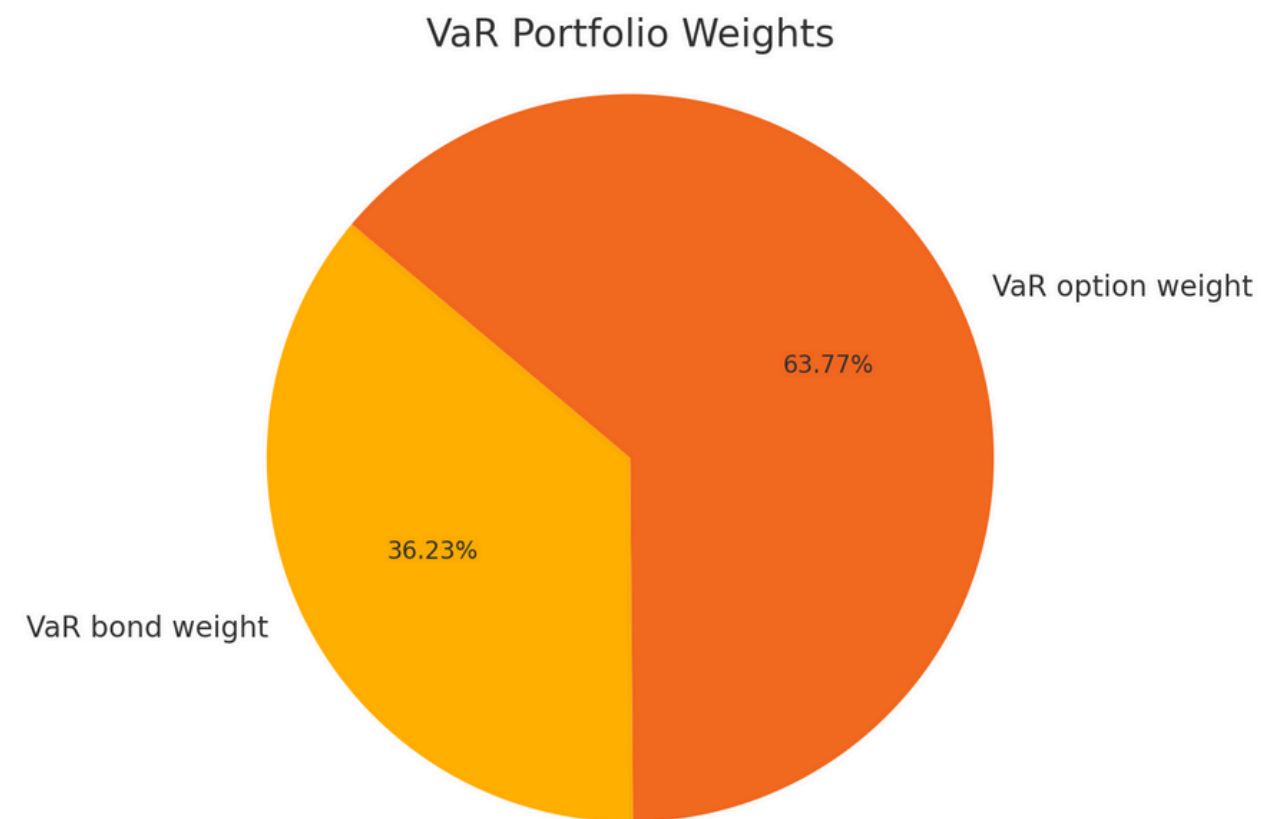
- Historical Simulation shows higher risk, capturing potential non-linear effects and tail risks.
- Delta-Normal (Cash-Flow Mapping) relies on a normal distribution and linear sensitivities, underestimating extreme risks.

Part III

Q1: Aggregate VaR and ES for Regulatory Reporting



Q1: Aggregate VaR and ES for Regulatory Reporting



Assumptions & Notes:

- FX risk considered implicitly in CHF conversion
- Correlation applied between equity and bond portfolios
- ES aggregation used a conservative additive method (sum of component ES)

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Thank

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